

The Theory of Endogenous Model Risk

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Abstract

This treatise develops a theory of endogenous model risk: the proposition that model risk is not confined to uncertainty about the correctness of an individual model, but may arise endogenously from the interaction between models, evaluation regimes, model selection, institutions, agents, and the subsequent adaptation of the system being modelled. The theory is deliberately higher-order. A model influences the evaluation environment in which it is assessed; the evaluation environment influences model selection and deployment; deployment changes behaviour and therefore the data-generating process; and the altered data-generating process feeds back into subsequent model evaluation.

The framework synthesises statistical epistemology, econometrics, finance, economics, game theory, control theory, physics, chemistry, biology, psychology, philosophy of science, political economy, international relations, warfare economics, computer science, algorithms, data science, artificial intelligence, and machine learning. The principal object is a higher-order model-risk field that combines predictive loss, model disagreement, regime instability, and evaluation distortion. The resulting supermodel is not asserted to be universally correct: it is itself admitted to the model space and therefore remains subject to model risk.

The theory extends the distinction between cross-sectional model unreliability and individual epistemic uncertainty developed in the statistical meta-theory of meta-theories, while drawing upon the Aggregation–Distortion framework for understanding information loss under aggregation. The objective is not to eliminate model uncertainty, but to model it explicitly, diversify it, and make its dynamics empirically estimable.

The treatise ends with “The End”

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1 Introduction

Scientific and economic inquiry rarely operates with a single model. Economists use structural, reduced-form, behavioural, rational-expectations, institutional, and agent-based models. Finance employs factor models, stochastic-volatility models, equilibrium models, term-structure models, and risk-neutral pricing

systems. Physics employs competing approximations and effective theories. Biology uses mechanistic, statistical, evolutionary, and systems-level representations. Artificial intelligence employs symbolic, statistical, neural, causal, and hybrid models.

Consequently, the relevant object of scientific risk is often not an individual model but a *model ecosystem*.

The conventional formulation of model risk is approximately:

$$\text{Model Risk} = \Pr(\text{selected model is inadequate}). \quad (1)$$

This is insufficient whenever the act of evaluation changes the environment being evaluated.

The present theory instead begins from the recursive structure

$$\text{Model} \longrightarrow \text{Evaluation} \longrightarrow \text{Selection} \longrightarrow \text{Deployment} \longrightarrow \text{Behaviour} \longrightarrow \text{Data} \longrightarrow \text{Evaluation}. \quad (2)$$

The central proposition is therefore:

$$\boxed{\text{Model performance may be endogenous to the evaluation regime.}} \quad (3)$$

This proposition connects naturally to Goodhart-type mechanisms, strategic adaptation, Lucas-type policy invariance problems, financial reflexivity, institutional selection, and the broader problem of reflexive epistemology [1, 2, 3, 10, 12].

The theory is deliberately reflexive. If the theory itself belongs to the admissible model space, it must be exposed to the same model-risk criterion as every other model. This follows the principle of reflexive non-exemption developed in the measure-theoretic meta-theory of meta-theories [11].

2 The Epistemological Foundation

2.1 Theories, models, and meta-models

Let $\mathcal{M}$ denote an admissible space of models. A model is understood broadly as a formal, semi-formal, or computational object that generates explanations, predictions, classifications, prescriptions, or inferential rules.

A meta-model is a model whose domain contains models themselves.

Thus,

$$M_k \in \mathcal{M}, \quad \mathcal{H}(\mathcal{M}) \in \mathcal{M}, \quad (4)$$

where $\mathcal{H}$ is a higher-order model that operates on a collection of lower-order models.

The possibility that a meta-model belongs to the same model space that it evaluates creates reflexivity. A framework that evaluates other frameworks cannot automatically exempt itself without restricting the domain of its own claim [10, 11].

2.2 Evaluation regimes

Correctness is not treated as a primitive absolute property. Let

$$E = (D, S, \kappa) \quad (5)$$

denote an evaluation regime, where D is the domain of application, S is the semantic interpretation standard, and κ is the criterion of adequacy.

The criterion may represent logical consistency, empirical adequacy, predictive calibration, explanatory power, normative coherence, or correspondence with a specified target.

Define

$$C_E(M) = \begin{cases} 1, & \text{if } M \text{ is correct relative to } E, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The measure-theoretic meta-theory emphasizes that the evaluation regime itself is a modelling choice and that the probability measure over admissible meta-theories is generally non-canonical [11].

2.3 The statistical fallibilist principle

A population-level statement about model reliability must not automatically be converted into an individual probability statement.

Let μ be a probability measure over $\mathcal{M}$ and let $\mathcal{I}_E$ denote the set of models incorrect under evaluation regime E .

The Half-Incorrectness principle is

$$\mu(\mathcal{I}_E) \geq \frac{1}{2}. \quad (7)$$

This is a cross-sectional statement. It does not imply that every fixed model has probability at least one half of being incorrect. The distinction between population-level unreliability and individual epistemic credence is essential for model-risk analysis [10, 11].

3 From Model Uncertainty to Endogenous Model Risk

3.1 The model ecosystem

Consider K competing models:

$$M_k : \quad y_t = f_k(x_t, \theta_k, s_t) + \varepsilon_{kt}, \quad k = 1, \dots, K. \quad (8)$$

Here s_t denotes a potentially latent regime.

Let

$$\hat{y}_{kt} = f_k(x_t, \hat{\theta}_{kt}, s_t) \quad (9)$$

be the prediction generated by model k .

Define the model-specific loss

$$L_{kt} = L(y_t, \hat{y}_{kt}). \quad (10)$$

A conventional selection rule is

$$k_t^* = \arg \min_k L_{kt}. \quad (11)$$

The present theory rejects the presumption that such selection is necessarily sufficient. Instead, the complete model ecosystem is retained.

3.2 Model disagreement

Define the ensemble prediction

$$\hat{y}_t^H = \sum_{k=1}^K w_{kt} \hat{y}_{kt}, \quad \sum_{k=1}^K w_{kt} = 1, \quad (12)$$

and define model disagreement as

$$D_t = \sum_{k=1}^K w_{kt} (\hat{y}_{kt} - \hat{y}_t^H)^2. \quad (13)$$

A low average prediction error accompanied by high disagreement is fundamentally different from low error accompanied by convergence across structurally different models.

Thus model disagreement is itself an observable or estimable component of model risk.

3.3 Evaluation distortion

The Aggregation–Distortion framework provides a useful foundation. If x is a vector of micro-level states and $A(x)$ is an aggregation operator, then information destroyed by aggregation can contribute to expected decision or inference loss [12].

A stylised representation is

$$E[L(\phi \circ A)] \geq E[L(\phi^*)] + I(x; x \mid A(x)), \quad (14)$$

where ϕ^* denotes the rule operating on the disaggregated information and the final term represents information lost through aggregation.

For model-risk purposes, define an evaluation-distortion measure

$$G_t = D(E_t, E_t^0), \quad (15)$$

where E_t^0 is a benchmark evaluation regime regarded as exogenous for the purpose of comparison.

4 Theory of Endogenous Model Risk

4.1 Definition

Definition 1 (Endogenous Model Risk). Endogenous model risk is the discrepancy between model performance under an exogenous evaluation regime and model performance under an evaluation regime that has itself been influenced by the models being evaluated.

Formally, let

$$L_{kt}^0 = L(y_t, \hat{y}_{kt}^0) \quad (16)$$

denote loss under an exogenous benchmark regime and let

$$L_{kt}^E = L(y_t, \hat{y}_{kt}^E) \quad (17)$$

denote loss under an endogenous regime. Then the model-specific endogenous risk component is

$$ER_{kt} = L_{kt}^E - L_{kt}^0. \quad (18)$$

The sign of ER_{kt} need not be fixed. Endogeneity may make apparent performance either better or worse.

4.2 Endogeneity of evaluation

Let

$$E_t = \mathcal{E}(\mathcal{M}_{t-1}, D_{t-1}, A_{t-1}, Z_t), \quad (19)$$

where A_{t-1} represents actions induced by previous model evaluations and Z_t contains exogenous environmental variables.

Then model performance satisfies

$$L_{kt} = L_k(y_t, x_t, E_t, \theta_k). \quad (20)$$

Therefore,

$$\frac{\partial L_{kt}}{\partial E_t} \neq 0 \quad (21)$$

is possible even when the model itself has not changed.

4.3 Proposition: endogenous evaluation

Proposition 1. If the evaluation regime depends on the prior model ecosystem and model performance depends on the evaluation regime, then model performance is dynamically endogenous.

Proof. Suppose

$$E_t = \mathcal{E}(\mathcal{M}_{t-1}, D_{t-1}) \quad (22)$$

and

$$L_{kt} = L_k(E_t, \theta_k, D_t). \quad (23)$$

Then a change in $\mathcal{M}_{t-1}$ can change E_t , while a change in E_t can change L_{kt} . Hence model performance is not a function solely of the structural parameters of M_k ; it is also a function of the endogenous evaluation process. Therefore model performance is dynamically endogenous. $\square$

5 The Model-Risk Supermodel

5.1 Definition

The higher-order model is defined by

$$\mathcal{H} : \mathcal{M} \longrightarrow \mathcal{R}, \quad (24)$$

where $\mathcal{R}$ is the space of model-risk states.

Define

$$R_t = \sum_{k=1}^K w_{kt} L_{kt} + \lambda_D D_t + \lambda_G G_t, \quad (25)$$

where $\lambda_D, \lambda_G \geq 0$.

The three terms represent expected model loss, cross-model disagreement, and evaluation distortion respectively.

Thus

$$\boxed{R_t = \text{Expected Model Loss} + \text{Model Disagreement} + \text{Evaluation Distortion.}} \quad (26)$$

This is the proposed *Model-Risk Field*.

5.2 Dynamic model weights

Let

$$Q_{kt} = L_{kt} + \lambda_D D_{kt} + \lambda_G G_{kt} + \lambda_S S_{kt}, \quad (27)$$

where S_{kt} measures structural instability.

The model weights evolve according to

$$w_{kt} = \frac{\exp(-\eta_t Q_{kt})}{\sum_{j=1}^K \exp(-\eta_t Q_{jt})}. \quad (28)$$

The temperature or learning parameter η_t can itself be dynamic:

$$\eta_t = \eta_0 + \eta_1 R_{t-1}. \quad (29)$$

This allows model selection intensity to respond to previously observed model risk.

6 Reflexive Higher-Order Modelling

The supermodel is not exempt from model risk.

Let

$$\mathcal{M}^{(0)} = \{M_1, \dots, M_K\} \quad (30)$$

denote first-order models.

Let

$$\mathcal{M}^{(1)} = \mathcal{H}(\mathcal{M}^{(0)}) \quad (31)$$

denote the first higher-order model.

A further meta-model is

$$\mathcal{M}^{(2)} = \mathcal{H}^{(2)}(\mathcal{M}^{(1)}). \quad (32)$$

Hence,

$$\mathcal{M}^{(0)} \longrightarrow \mathcal{M}^{(1)} \longrightarrow \mathcal{M}^{(2)} \longrightarrow \dots \quad (33)$$

The hierarchy need not be infinite in practice. A finite truncation can be chosen according to computational, informational, or economic constraints.

Theorem 1 (Reflexive Non-Exemption). If the endogenous model-risk supermodel belongs to the admissible model space, then it is subject to the same model-risk criterion as the models it evaluates.

Proof. Let $\mathcal{H} \in \mathcal{M}$. Since the model-risk criterion is defined on $\mathcal{M}$, it applies to every member of $\mathcal{M}$, including $\mathcal{H}$. Exempting $\mathcal{H}$ would require either restricting the domain of the criterion or denying that $\mathcal{H} \in \mathcal{M}$. Therefore $\mathcal{H}$ cannot claim structural exemption from its own model-risk criterion. $\square$

This is the model-risk analogue of reflexive non-exemption in the statistical and measure-theoretic meta-theories of meta-theories [10, 11].

7 Connections with Economics and Finance

7.1 Economics

Economic agents respond to models, forecasts, policies, and institutions. Consequently, an economic model can become part of the environment it seeks to describe.

A policy rule can therefore be represented as

$$a_t = g(\hat{\theta}_t, \hat{y}_t), \quad (34)$$

while private agents respond strategically:

$$x_{t+1} = F(x_t, a_t, E_t). \quad (35)$$

The model changes the policy, the policy changes behaviour, and behaviour changes the data-generating process.

This provides a formal bridge between endogenous model risk and rational expectations, policy invariance, Goodhart's Law, and Lucas's critique [2, 1].

7.2 Finance

In finance, model risk appears in derivatives pricing, portfolio allocation, risk management, stress testing, volatility estimation, and asset-pricing models.

Suppose asset returns satisfy

$$r_{t+1} = \mu_t + \sigma_t \varepsilon_{t+1}. \quad (36)$$

Different volatility models imply different risk measures:

$$VaR_{\alpha,t}^{(k)} = q_{\alpha}(r_{t+1} \mid M_k, \mathcal{F}_t). \quad (37)$$

Model disagreement can then be represented by

$$D_t^{VaR} = \sum_{k=1}^K w_{kt} \left(VaR_{\alpha,t}^{(k)} - \overline{VaR}_{\alpha,t} \right)^2. \quad (38)$$

The resulting risk measure should therefore distinguish parameter uncertainty, residual uncertainty, regime uncertainty, and model uncertainty.

7.3 Saturation and structural models

The Standard Nuclear oliGARCHy provides an instructive example of the distinction between mathematical model claims and empirical claims. In the supplied saturation paper, a deterministic SNoG construction generates a saturation trajectory and therefore produces zero residual sum of squares and $R^2 = 1$ within the stipulated noiseless model [13].

The result is mathematical conditional on the model’s premises rather than an assertion that arbitrary empirical data must satisfy $R^2 = 1$.

This distinction is essential to endogenous model-risk theory: mathematical certainty conditional on a model does not remove uncertainty concerning whether the model’s premises apply to a particular empirical system.

8 Connections with Physics and Other Natural Sciences

Physics frequently uses multiple representations of the same phenomenon: microscopic models, effective theories, continuum approximations, stochastic processes, and computational simulations.

Let

$$M_{\text{micro}}, \quad M_{\text{meso}}, \quad M_{\text{macro}} \quad (39)$$

represent models operating at different scales.

An aggregation operator

$$A : X_{\text{micro}} \longrightarrow X_{\text{macro}} \quad (40)$$

may discard information.

This is structurally analogous to the Aggregation–Distortion framework [12].

Chemistry presents a similar hierarchy between molecular mechanisms, reaction networks, thermodynamic approximations, and empirical rate laws.

Biology adds another layer through hierarchical regulation:

$$\text{gene} \rightarrow \text{cell} \rightarrow \text{organ} \rightarrow \text{organism} \rightarrow \text{population}. \quad (41)$$

At each level, a model may be accurate within a particular regime while failing outside it.

The present theory therefore treats model risk as a general property of hierarchical scientific representation rather than an exclusively financial phenomenon.

9 Psychology and Psychiatry

Psychological systems introduce an especially important reflexive mechanism: agents may alter behaviour when they learn the model used to predict them.

Let

$$a_t = \pi(x_t, \widehat{M}_t), \quad (42)$$

where $\widehat{M}_t$ represents the agent’s belief concerning the model used by the evaluator.

Then

$$\frac{\partial a_t}{\partial \widehat{M}_t} \neq 0 \quad (43)$$

creates direct model-induced behavioural adaptation.

Psychiatric models similarly operate under regime-specific diagnostic criteria, institutional incentives, measurement conventions, and changing classifications. The present theory does not assert clinical laws; rather, it identifies a general methodological problem: an evaluation framework may alter the object it classifies.

The distinction between objective accuracy and subjective confidence is already central to statistical epistemology [9].

10 Philosophy of Science

The theory sits at the intersection of falsifiability, Bayesian epistemology, model pluralism, underdetermination, reflexivity, and scientific realism.

Falsification remains valuable at the individual-model level. It is not, however, the entire architecture of scientific risk management.

A sophisticated model ecosystem may contain

$$M_1, M_2, \dots, M_K \quad (44)$$

without requiring one model to be declared universally correct.

The epistemological objective therefore becomes

$$\text{model selection} \longrightarrow \text{model diversification} \longrightarrow \text{model-risk estimation}. \quad (45)$$

The framework consequently treats model uncertainty as an object to be measured rather than merely a defect to be eliminated.

11 Political Science, Geopolitics, and Diplomacy

Political and geopolitical models are particularly susceptible to reflexivity because actors may respond to forecasts and strategic assessments.

Let i index strategic actors and let

$$u_i(a_i, a_{-i}; M) \quad (46)$$

denote utility under a model M .

If actors observe or infer M , then their best-response correspondence becomes

$$BR_i(M) = \arg \max_{a_i} u_i(a_i, a_{-i}; M). \quad (47)$$

The model therefore enters the strategic environment.

Diplomatic signalling, deterrence, sanctions, alliance formation, military planning, and international economic policy can all generate such feedback.

The relevant distinction is between

$$\text{prediction without strategic reaction} \quad (48)$$

and

$$\text{prediction with strategic reaction to prediction}. \quad (49)$$

The latter is intrinsically reflexive.

12 Welfare Economics and Warfare Economics

A social planner may optimize

$$W_t = W(c_t, l_t, g_t, r_t), \quad (50)$$

while the estimated welfare function itself determines policy.

If policy changes behaviour, then

$$\widehat{W}_{t+1} \neq W_{t+1}(\widehat{\theta}_t) \quad (51)$$

in general.

Warfare economics introduces even stronger strategic feedback. Military actors adapt to threat assessments, while threat assessments adapt to observed military behaviour.

A stylised strategic state transition is

$$x_{t+1} = F(x_t, u_t^A, u_t^B), \quad (52)$$

where the actions of opposing actors depend upon their assessments:

$$u_t^A = \pi_A(\widehat{M}_t^A), \quad u_t^B = \pi_B(\widehat{M}_t^B). \quad (53)$$

Consequently, model risk can become strategic risk.

The SNoG framework provides an example in which deterrence is itself embedded in the structural dynamics of the model [13]. The present theory generalises the methodological lesson without assuming that every strategic system has the same dynamics.

13 Computer Science, Algorithms, and Artificial Intelligence

13.1 Graph representation

Represent the model ecosystem as a directed graph

$$G = (V, E), \quad (54)$$

where each node represents a model and each edge represents an evaluation, dependence, comparison, or transformation.

The graph may contain cycles:

$$M_1 \rightarrow E \rightarrow M_2 \rightarrow E \rightarrow M_1. \quad (55)$$

Cycles are not automatically errors. They are the computational representation of reflexivity.

13.2 Model-risk algorithm

```

EndogenousModelRisk(M, D, E):
  initialise model weights w
  initialise risk state R

  repeat:
    for each model M_k in M:
      fit M_k to D
      generate predictions
      compute predictive loss L_k
      compute disagreement D_k
      compute structural instability S_k

    estimate evaluation distortion G

  compute model-risk state
  R <- weighted loss
    + disagreement penalty
    + distortion penalty

  update model weights
  w_k <- exp(-eta * Q_k)
  normalise w

```

```

    update evaluation regime
      E <- EvaluationUpdate(M, D, w, R)

    update data-generating environment
      D <- EnvironmentUpdate(E, M, D)

until convergence or stopping criterion

return M, w, R, E

```

The crucial difference from ordinary ensemble learning is the feedback from evaluation to the data-generating environment.

13.3 Machine learning

Let training loss be

$$L_{\text{train}}(\theta) \quad (56)$$

and generalisation loss be

$$L_{\text{gen}}(\theta). \quad (57)$$

A paradigm-dependent training process may minimize

$$L_{\text{train}} + \Omega(\Pi), \quad (58)$$

where Π represents the prevailing paradigm.

If the true data-generating process lies outside Π , excessive conformity can increase systematic misspecification. The supplied *Lies, Damned Lies, and Institutions* framework interprets this as an analogue of paradigm overfitting [12].

The endogenous model-risk framework generalises this intuition:

$$\text{training} \rightarrow \text{selection} \rightarrow \text{deployment} \rightarrow \text{adaptation} \rightarrow \text{new training distribution}. \quad (59)$$

This is a central concern in reinforcement learning, online learning, adaptive control, and multi-agent systems.

14 Econometric Method

14.1 Competing-model panel

Estimate K structural or statistical models and retain their full loss histories:

$$\mathbf{L}_t = (L_{1t}, L_{2t}, \dots, L_{Kt}). \quad (60)$$

The model-loss panel becomes the primary econometric object.

14.2 Regime switching

Let

$$s_t \in \{1, \dots, S\} \quad (61)$$

follow a Markov process with transition probabilities

$$P(s_t = j \mid s_{t-1} = i) = p_{ij}. \quad (62)$$

Model performance is allowed to vary across regimes:

$$E[L_{kt} \mid s_t = s] = \mu_{ks}. \quad (63)$$

14.3 Endogenous evaluation equation

Let the evaluation weights satisfy

$$w_{kt} = g_k(\mathbf{L}_{t-1}, D_{t-1}, S_{t-1}, Z_t). \quad (64)$$

This equation directly tests whether model selection is itself endogenous.

14.4 Model-risk equation

The central empirical equation is

$$R_t = \alpha + \rho R_{t-1} + \beta' X_t + \gamma' Z_t + \delta D_t + u_t. \quad (65)$$

A central hypothesis is

$$H_0 : \frac{\partial E[L_{kt}]}{\partial E_t} = 0, \quad (66)$$

against

$$H_1 : \frac{\partial E[L_{kt}]}{\partial E_t} \neq 0. \quad (67)$$

Identification requires careful treatment of simultaneity, omitted variables, measurement error, and reverse causality. Possible strategies include instrumental variables, structural state-space models, natural experiments, randomized evaluation changes where feasible, and externally imposed regime shifts.

14.5 Model-Risk Endogeneity Ratio

Define

$$MER = \frac{\text{Var}(\hat{L}_{kt}^E - \hat{L}_{kt}^{E_0})}{\text{Var}(\hat{L}_{kt}^{E_0})}. \quad (68)$$

The statistic is a diagnostic rather than a universal measure. Its sampling distribution and identification properties must be established separately for each application.

15 Statistical Diagnostics

The diagnostic suite should include predictive error, calibration, parameter stability, regime stability, model disagreement, and sensitivity to the evaluation regime.

Useful quantities include

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (69)$$

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (70)$$

and

$$R^2 = 1 - \frac{SSR}{SST}. \quad (71)$$

However, no single goodness-of-fit statistic should be treated as a universal measure of model correctness.

The SNoG saturation result illustrates why. When the data are mathematically generated by the fitted saturation function, $SSR = 0$ and $R^2 = 1$ follow deductively within the stipulated construction [13]. This is not equivalent to the claim that an arbitrary empirical dataset must achieve $R^2 = 1$.

16 Cross-Disciplinary Interpretation

Table 1: Interpretations of endogenous model risk across fields.

Field	Model object	Endogeneity mechanism
Economics	Structural model	Policy and behavioural response
Finance	Pricing/risk model	Market adaptation
Econometrics	Estimator/model class	Specification feedback
Physics	Effective theory	Scale-dependent approximation
Chemistry	Reaction model	Intervention-induced dynamics
Biology	Regulatory model	Evolution and adaptation
Psychology	Behavioural model	Reflexive subject response
Philosophy	Meta-theory	Self-reference
Political science	Institutional model	Strategic adaptation
Geopolitics	Strategic model	Adversarial response
Warfare economics	Conflict model	Deterrence and counter-adaptation
Computer science	Model graph	Recursive evaluation
AI/ML	Learning system	Distribution shift and adaptation
Data science	Pipeline	Selection-induced bias

These are structural analogies and methodological extensions rather than claims that the same mathematical law governs every discipline.

17 A General Theorem of Model-Risk Recursion

Theorem 2 (Model-Risk Recursion). Suppose that:

1. the model ecosystem influences the evaluation regime;
2. the evaluation regime influences model selection;
3. model selection influences the environment;
4. the environment influences subsequent observations.

Then model risk is dynamically recursive.

Proof. Let

$$E_t = \mathcal{E}(\mathcal{M}_{t-1}, D_{t-1}), \quad (72)$$

let selection satisfy

$$w_t = \mathcal{W}(E_t, \mathcal{M}_{t-1}), \quad (73)$$

and let the environment evolve according to

$$D_t = \mathcal{D}(D_{t-1}, w_t, E_t). \quad (74)$$

Then

$$D_t = \mathcal{D}[D_{t-1}, \mathcal{W}(\mathcal{E}(\mathcal{M}_{t-1}, D_{t-1}), \mathcal{M}_{t-1}), \mathcal{E}(\mathcal{M}_{t-1}, D_{t-1})]. \quad (75)$$

Thus the current data-generating environment depends upon prior models through the evaluation and selection mechanisms. Consequently, future model performance depends upon prior model selection. Model risk is therefore recursive. $\square$

18 Policy, Institutional, and Strategic Implications

The theory implies that institutions should not treat model validation as a one-time certification event. Instead, a model-risk architecture should preserve:

model plurality+independent evaluation+regime monitoring+model disagreement+reflexive validation. (76)

This does not imply that decentralisation is universally optimal or that peer review is universally harmful. The supplied institutional analysis argues that gatekeeping can destroy information when evaluation is strongly conditioned on prevailing paradigms, while an epistemic-market architecture can preserve competing claims [12]. These propositions themselves remain empirical claims subject to evaluation.

In finance and macroeconomics, the practical implication is to avoid confusing a successful historical fit with structural invariance.

In strategic environments, the implication is stronger: publishing or deploying a model can itself alter strategic behaviour.

In artificial intelligence, a benchmark can become a target, causing the system to optimize for the benchmark rather than the underlying objective.

Thus:

$$\boxed{\text{A model may change the world that validates the model.}} \quad (77)$$

19 Relation to the Meta-Theory of Meta-Theories

The present theory is naturally interpreted as a lower-order application of reflexive meta-theoretical reasoning.

The axiomatic meta-theory assigns a prior probability of correctness of

$$P(C) = \frac{1}{e} \quad (78)$$

under its stipulated Exponential Correctness Principle [9].

The later statistical formulation instead imposes

$$P(I(M) = 1) \geq \frac{1}{2} \quad (79)$$

at the population level and explicitly distinguishes this from individual epistemic credence [10].

The measure-theoretic formulation replaces unrestricted “all meta-theories” language with an admissible measurable space and an evaluation regime [11].

The endogenous model-risk theory adopts the same reflexive discipline:

$$\boxed{\text{The model of model risk is itself a model.}} \quad (80)$$

Therefore it cannot claim immunity from model risk.

20 Research Programme

The theory suggests a sequence of increasingly demanding empirical and computational projects:

1. construct competing structural models for a common phenomenon;
2. estimate their predictive and structural losses;
3. measure cross-model disagreement;
4. identify changes in evaluation regimes;
5. estimate endogenous model weights;
6. estimate the Model-Risk Field;

7. test regime dependence;
8. perform Monte Carlo experiments;
9. apply the framework to financial and macroeconomic data;
10. extend the framework to multi-agent reinforcement learning;
11. construct higher-order model-risk models;
12. evaluate the model-risk model itself.

The final step is not optional.

$$\mathcal{H} \in \mathcal{M} \implies R(\mathcal{H}) \text{ must be estimable.} \quad (81)$$

21 Conclusion

The Theory of Endogenous Model Risk begins with a simple observation: models do not necessarily remain external to the systems they describe.

Models influence evaluations. Evaluations influence selection. Selection influences behaviour. Behaviour changes the data. New data change evaluation. Consequently, model risk can become endogenous.

The appropriate response is not necessarily to search for a single universally correct model. A more robust architecture retains alternative models, measures disagreement, estimates regime dependence, models evaluation distortion, and constructs higher-order representations of the model ecosystem itself.

The resulting structure is

$$\boxed{\text{Models} \rightarrow \text{Evaluation} \rightarrow \text{Selection} \rightarrow \text{Adaptation} \rightarrow \text{Data} \rightarrow \text{Models}.} \quad (82)$$

The central theoretical object is the Model-Risk Field,

$$R_t = \sum_{k=1}^K w_{kt} L_{kt} + \lambda_D D_t + \lambda_G G_t. \quad (83)$$

Its purpose is not to provide a final answer to model uncertainty. Its purpose is to make model uncertainty itself an explicit object of mathematical, statistical, econometric, computational, and institutional analysis.

The theory therefore ends where higher-order modelling begins:

$$\boxed{\text{A model of model risk is itself exposed to model risk.}} \quad (84)$$

That reflexive condition is not a defect. It is the defining property of the theory.

22 Figures

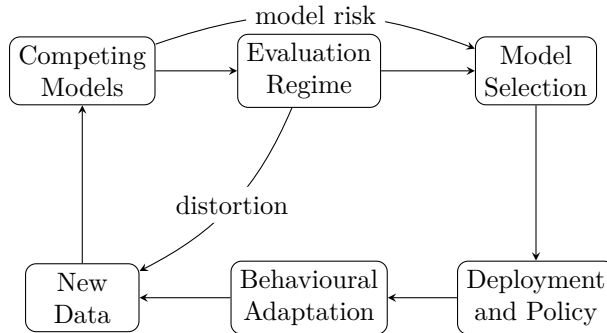


Figure 1: The recursive architecture of endogenous model risk.

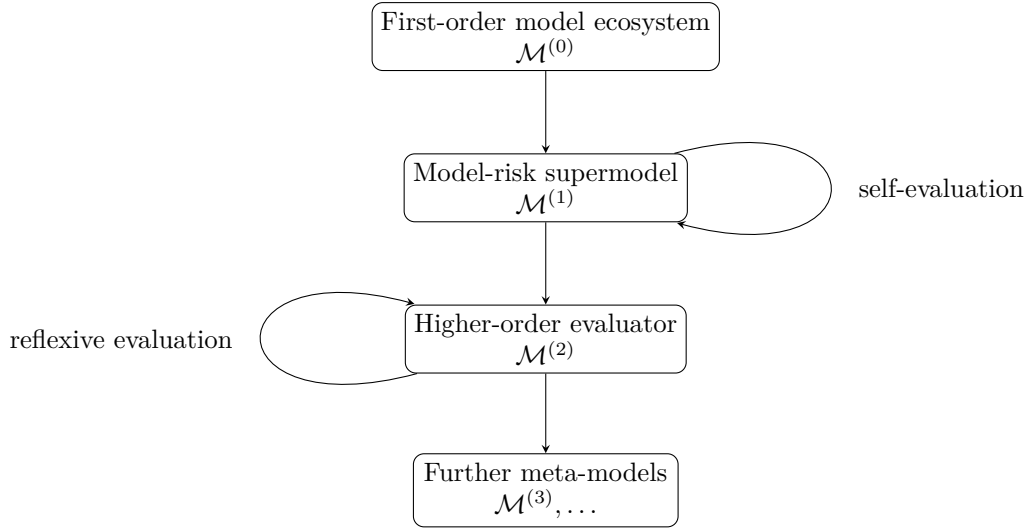


Figure 2: Higher-order model-risk recursion and reflexive non-exemption.

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Glossary

Admissible Model Space

The set of models satisfying the syntactic and semantic requirements of a specified modelling language.

Aggregation Distortion

The loss of information and consequent increase in inference or decision loss caused by mapping a richer state space into an aggregated representation.

Endogenous Model Risk

The discrepancy between model performance under an exogenous evaluation regime and performance under an evaluation regime influenced by the models themselves.

Evaluation Regime

A specification of the domain, semantic interpretation, and criterion used to evaluate a model.

Evaluation Distortion

The component of model risk arising because the evaluation mechanism differs from an exogenous benchmark.

Model Disagreement

The dispersion of predictions, parameters, or risk estimates across competing models.

Model Ecosystem

A collection of competing models together with the evaluation, selection, deployment, and feedback mechanisms connecting them.

Model-Risk Field

The higher-order state variable combining expected model loss, model disagreement, and evaluation distortion.

Model-Risk Endogeneity Ratio

A diagnostic comparing variation in model performance under endogenous and benchmark evaluation regimes.

Meta-Model

A model whose objects include other models.

Reflexive Non-Exemption

The principle that a meta-model belonging to its own admissible model space is subject to the same evaluation criteria as other models.

Regime

A state of the environment in which the parameters or structural relationships governing model performance differ from those of other states.

Supermodel

A higher-order model that represents, combines, evaluates, or governs a collection of lower-order models.

Model Pluralism

The deliberate retention of multiple competing models rather than assuming that one model is universally correct.

Goodhart Mechanism

The phenomenon in which a statistical regularity changes when the regularity itself becomes a target of optimization or policy.

Model Risk

The possibility that a model, its assumptions, its implementation, its evaluation, or its use produces materially inadequate inference or decisions.

The End